

Figure 1.24 An SVG circle with fill style determined by its type and its opacity and stroke settings determined by its membership in the `tentative` class.

reload the page and take advantage of the helper function `d3.classed()`, which allows you to add or remove a class from the classes in an element:

```
d3.select("circle").classed("active", true);
```

By using `.classed()`, you don't overwrite the existing attribute, but rather append or remove the named class from the list. You can see the results of two classes with conflicting styles defined. The active style overwrites the tentative style because it occurs later in the style sheet. Another rule to remember is that more specific rules overwrite more general rules. There's more to CSS, but this book won't go into that.

By defining style in your style sheet and changing appearance based on class membership, you create code that's more maintainable and readable. You'll need to use inline styles to set the graphical appearance of a set of elements to a variety of different values, for example, changing the fill color to correspond to a color ramp based on the data bound to that set of elements. You'll see that functionality in action later when you deal with bound data. But as a general rule, setting inline styles should only be used when you can't use traditional classes and states defined in a style sheet.

1.3.5 JavaScript

D3, like many information visualization libraries in JavaScript, provides functions to abstract the process of creating and modifying web page elements. On top of that, it provides mechanisms to link data and web page elements in a way that makes the drawing and updating of these SVG elements reusable and maintainable. But these mechanisms are also applicable to more traditional HTML elements like paragraphs and divs.

As a result, a web application written in D3 can accomplish much of the UI functionality that users expect without relying on libraries like jQuery. This is because the latest version of JavaScript has built-in functionality that used to be available only with jQuery. If you read the solutions on Stack Overflow, you may think that being a JavaScript developer requires being a jQuery developer, but unless you're developing for an audience that uses a browser that won't support the key features of D3, or you need to use a plugin that requires jQuery, then you might just as easily write the same functionality in JavaScript.

When writing JavaScript with D3, you should familiarize yourself with two subjects: method chaining and arrays.